

RETRACTED (2026-05-16)

*The main theorem of this note is **false**. The proof of Lemma 6 (Case 2b) silently assumed that the FPS monotonicity of $f_{2b}(\eta, \delta)$ in η on $[1/2, \eta_b(\delta))$ holds at every $\delta > 1$. In fact it holds only for $\delta \geq -3 + \sqrt{17} \approx 1.12311$; at the proposed $\delta_1 \approx 1.114907$, $f_{2b}^{\max}(\delta_1) \approx 0.5654 > 9/16$, so $F(\delta_1) > F(9/8) = 9/16$ and no improvement is obtained.*

*The corrected analysis shows that FPS's choice $\delta = 9/8$ is structurally forced and that $F(\delta)$ attains its minimum at this value, with $F(9/8) = 9/16$. See the companion note **sharpness_9_8.tex** for the corrected statement.*

The original (false) draft is preserved below for historical record.

A tighter constant in the chromatic-index lemma of Fox–Pach–Suk towards Albertson’s conjecture

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Abstract

Fox, Pach and Suk (SoCG 2025) recently proved Albertson’s conjecture $\text{cr}(G) \geq \text{cr}(K_k)$ for every k -chromatic graph G on at most $1.4(k-1)$ vertices. The central technical input is their Lemma 2.3, which bounds the chromatic index of an auxiliary multigraph by $(9/16 + o(1))k$. The constant $9/16$ in turn arises from a three-case optimisation (their Claim 3.7) parameterised by a degree threshold $d := \delta k$, set to $\delta = 9/8$. We observe that this choice of δ is not the analytic optimum of Claim 3.7’s case-bound function. Reoptimising over δ replaces $9/16$ by

$$F^* = -\frac{1}{2} + \frac{2}{\sqrt{3}} \cos\left(\frac{1}{3} \arccos \frac{3\sqrt{3}}{16}\right) \approx 0.557454,$$

the unique real root in $(\frac{1}{2}, \frac{5}{8})$ of $4x^3 + 6x^2 - x - 2 = 0$. Every other ingredient of Fox–Pach–Suk’s argument is checked to be robust to this change. The resulting tightening of Lemma 2.3 is modest in absolute terms ($\approx 0.9\%$) but follows immediately from their proof.

1 Introduction

Albertson conjectured [?] that for every $k \geq 1$ and every graph G with chromatic number $\chi(G) \geq k$,

$$\text{cr}(G) \geq \text{cr}(K_k), \tag{1}$$

where $\text{cr}(\cdot)$ denotes the crossing number. The conjecture is known for $k \leq 24$, with substantial recent progress: Albertson, Cranston and Fox [?] proved the case $k \leq 12$; Barát and Tóth [?] extended it to $k \leq 16$; Ackerman [?] to $k \leq 18$; and Cranston [?] to $k \leq 24$, with strong restrictions on the possible counterexamples at $k \in \{25, 26\}$.

Fox, Pach and Suk [?] prove (??) for every k -chromatic graph on at most $1.4(k-1)$ vertices (and asymptotically on $(1.64 - o(1))k$ vertices) by a clever weak-immersion argument. The crucial technical step is the following bound on the chromatic index of an auxiliary multigraph H constructed from G via a semi-random selection procedure (see ?? for the precise setup):

$$\chi'(H_i) \leq (c_{\text{FPS}} + o(1))k_i, \quad c_{\text{FPS}} = \frac{9}{16}. \tag{2}$$

This is their Lemma 2.3. They derive the constant $c_{\text{FPS}} = 9/16$ from a three-case optimisation parameterised by a single free real *degree threshold* δ ; throughout [?] they fix $\delta = 9/8$ from the start.

In this note we observe that the choice $\delta = 9/8$ in [?] is suboptimal for the case bound. Re-optimising the three-case bound in their Claim 3.7 with δ as a free parameter yields a smaller constant.

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Theorem 1. *Let F^* be the unique real root in $(\frac{1}{2}, \frac{5}{8})$ of $4x^3 + 6x^2 - x - 2 = 0$. Then (??) holds with c_{FPS} replaced by F^* .*

Equivalently, c_{FPS} can be replaced by

$$F^* = -\frac{1}{2} + \frac{2}{\sqrt{3}} \cos\left(\frac{1}{3} \arccos \frac{3\sqrt{3}}{16}\right) = 0.5574537707\dots$$

Since $F^* < \frac{9}{16} = 0.5625$, this is a strict improvement over [?, Lem. 2.3]. The size of the improvement is $\frac{9}{16} - F^* \approx 5.05 \cdot 10^{-3}$, or about 0.9%. The optimal degree threshold realising F^* is

$$\delta_1 = -1 + \frac{4}{\sqrt{3}} \cos\left(\frac{1}{3} \arccos \frac{3\sqrt{3}}{16}\right) = 1.1149075414767\dots,$$

the unique real root in $(1, 5/4)$ of the irreducible cubic

$$\delta^3 + 3\delta^2 - \delta - 4 = 0. \tag{3}$$

The improvement is intentionally small in scope: we change exactly one numerical constant inside Fox–Pach–Suk’s proof of their Lemma 2.3 and verify (??) that none of the other ingredients of the proof is affected. Whether F^* itself is sharp – or whether a more substantive improvement of (??) is possible by attacking the underlying Vizing–Gupta input – remains open (??).

The downstream impact on the vertex-bound coefficients of [?, Thm. 1.2] is straightforward but bookkeeping-heavy; we isolate it as ??.

2 Reconstruction of [?], Claim 3.7

We follow the notation of [?] throughout. Let G be a k -chromatic graph and write $\{V_1, \dots, V_r\}$ for the parts of the Gallai decomposition of G , with $G[V_i]$ being k_i -critical and $n_i := |V_i| \geq 2k_i - 1$. Fix an index i and drop subscripts: set $n = n_i$, $k = k_i$, $V = V_i$, and so on.

Construction of H ([?, §3]). Choose a real parameter $\delta > 1$ and set $d := \delta k$. Let $U(d) \subseteq V$ be the set of vertices of degree at least d , capped at $k - \varphi(k)$, where $\varphi(k) := k^{0.9}$ is the slack term of [?, Footnote 1, p. 6]. Add to $U(d)$ a uniformly random set of $k - \ell$ vertices from $V \setminus U(d)$, where $\ell = |U(d)|$, to form U . After a deterministic edge-swap pre-processing ([?, p. 7–8]) and an additional uniform-random completion of the maps $f_u: U \setminus (N(u) \cup \{u\}) \rightarrow N(u) \cap W$, one obtains the multigraph H on $W := V \setminus U$. By Vizing–Gupta [?, Lem. 2.2(ii)],

$$\chi'(H) \leq \Delta(H) + \mu(H). \tag{4}$$

The bound (??) reduces ([?, §3]) to two high-probability estimates:

$$\textbf{Proposition 3.3.} \quad \Delta(H) \leq (F(\delta) + o(1)) k, \tag{5}$$

$$\textbf{Proposition 3.4.} \quad \mu(H) = o(k), \tag{6}$$

where $F(\delta)$ is the upper-bound function whose value $F(9/8) = 9/16$ is established by [?, Claim 3.7].

The optimisation ([?, Claim 3.7]). The function $F(\delta)$ is the supremum of the asymptotic expression

$$f(\alpha, \beta, \gamma; \delta) = \gamma - \alpha \frac{\delta - 1}{\delta - \gamma}, \quad (7)$$

over the feasible region

$$0 \leq \alpha \leq \beta \leq 1, \quad 0 \leq \gamma \leq \alpha + (\delta - \alpha) \frac{1 - \beta}{2 - \beta}. \quad (8)$$

Here α, β, γ are the normalisations $\alpha = \ell_w/k$, $\beta = \ell/k$, $\gamma = |U_w|/k$, where ℓ_w is the number of $U(d)$ -neighbours of a target vertex $w \in W$ and $U_w := N(w) \cap U$.

Following [?, Proof of Claim 3.7], the supremum splits into three cases:

Case 1. $\alpha = 0$.

Case 2a. The upper constraint on γ in (??) is tight, and the stationary point of f in α is interior to $[0, \beta]$.

Case 2b. The upper constraint on γ is tight, but the stationary point falls outside $[0, \beta]$, forcing $\alpha = \beta$.

Reparametrising $\eta := 1 - (1 - \beta)/(2 - \beta) \in [\frac{1}{2}, 1]$ (equivalently $\beta = 2 - 1/\eta$), Case 2 has $\gamma = \eta\alpha + (1 - \eta)\delta$.

3 The optimisation with δ free

We compute the maximum of f in each case as an explicit function of δ . All three formulas reduce to simple expressions and lead to the cubic (??).

Lemma 2 (Case 1). *For every $\delta \geq 1$, the maximum of f on the feasible set restricted to $\alpha = 0$ is*

$$f_1(\delta) = \frac{\delta}{2},$$

attained at $\beta = 0, \gamma = \delta/2$.

Proof. With $\alpha = 0$, (??) gives $f = \gamma$, and the upper bound on γ in (??) simplifies to $\gamma \leq \delta(1 - \beta)/(2 - \beta)$. This is a decreasing function of $\beta \in [0, 1]$ with derivative $-1/(2 - \beta)^2 < 0$, hence its maximum over $\beta \in [0, 1]$ is at $\beta = 0$, with value $\delta/2$. $\square$

Lemma 3 (Case 2 stationary point and 2a/2b boundary). *For every $\delta > 1$ and $\eta \in [\frac{1}{2}, 1]$, the unique stationary point in α of f on the Case-2 line $\gamma = \eta\alpha + (1 - \eta)\delta$ is*

$$\alpha^*(\eta, \delta) = \delta - \frac{\sqrt{\delta(\delta - 1)}}{\eta}. \quad (9)$$

This stationary point equals $\beta(\eta) = 2 - 1/\eta$ exactly when

$$\eta = \eta_b(\delta) := \frac{\sqrt{\delta(\delta - 1)} - 1}{\delta - 2}. \quad (10)$$

The Case 2a regime is $\eta \in [\eta_b(\delta), 1]$ and Case 2b is $\eta \in [\frac{1}{2}, \eta_b(\delta))$. At $\delta = 9/8$ this recovers [?, (p. 10)]: $\alpha^ = (9 - 3/\eta)/8$ and $\eta_b = 5/7$.*

Proof. Direct differentiation of $f(\alpha, \cdot, \eta\alpha + (1 - \eta)\delta; \delta)$ in α and setting the result to zero yields a quadratic in α with the two roots $\delta \pm \sqrt{\delta(\delta - 1)}/\eta$. Only $\alpha^* = \delta - \sqrt{\delta(\delta - 1)}/\eta$ lies in $[0, \beta]$ (at $\delta = 9/8$ and $\eta = 3/4$ it gives $\alpha^* = 5/8$, while the other root gives $\alpha = 5/3 > 1$). Equating $\alpha^*(\eta, \delta) = 2 - 1/\eta$ and solving for η gives (??). The substitutions at $\delta = 9/8$ are elementary. $\square$

Lemma 4 (Case 2a). *For every $\delta > 1$, f restricted to Case 2a is strictly decreasing in η on $[\eta_b(\delta), 1]$; its maximum is therefore attained at $\eta = \eta_b(\delta)$. Setting $\alpha = \alpha^*(\eta_b(\delta), \delta)$ and simplifying,*

$$f_{2a}^{\max}(\delta) = \frac{-\delta^{5/2} - \delta^{3/2} + 2\sqrt{\delta} + (\delta^2 + 2\delta - 2)\sqrt{\delta - 1}}{-\delta^{3/2} + \sqrt{\delta} + \sqrt{\delta - 1}}. \quad (11)$$

At $\delta = 9/8$, $f_{2a}^{\max}(9/8) = 11/20$, matching [?, p. 10].

Remark 5. The closed form (??) is what one obtains by substituting the stationary point (??) into the Case-2 expression $\gamma = \eta\alpha + (1 - \eta)\delta$ inside (??), then setting $\eta = \eta_b(\delta)$. The function is smooth on $\delta > 1$, tends to 1 as $\delta \rightarrow 1^+$, has a unique minimum near $\delta \approx 1.20$ with value ≈ 0.534 , and tends to $+\infty$ as $\delta \rightarrow +\infty$.

Lemma 6 (Case 2b). *For every $\delta > 1$, the maximum of f over Case 2b is attained at $\eta = \frac{1}{2}$ (so $\alpha = \beta = 0$) and equals*

$$f_{2b}^{\max}(\delta) = \frac{\delta}{2}.$$

Proof. At $\eta = \frac{1}{2}$, $\alpha = \beta = 2 - 1/\eta = 0$; from $\gamma = \eta\alpha + (1 - \eta)\delta = \delta/2$ and $f = \gamma - \alpha(\delta - 1)/(\delta - \gamma)$, $f = \delta/2$ since $\alpha = 0$.

The verification that $\eta = 1/2$ is the maximum (i.e. that f is monotone decreasing in η on $[\frac{1}{2}, \eta_b(\delta))$ under the substitution $\alpha = \beta = 2 - 1/\eta$) is a one-variable calculus check; at $\delta = 9/8$ it matches [?, p. 10] verbatim. $\square$

Proposition 7. *For every $\delta > 1$,*

$$F(\delta) = \max(f_1(\delta), f_{2a}^{\max}(\delta), f_{2b}^{\max}(\delta)) = \max(\frac{\delta}{2}, f_{2a}^{\max}(\delta)).$$

At $\delta = 9/8$, $f_{2a}^{\max}(9/8) = 11/20 < \delta/2 = 9/16$, so $F(9/8) = 9/16$, matching [?, Claim 3.7].

Proof. Lemmas ??, ?? give $f_1 = f_{2b}^{\max} = \delta/2$. $\square$

Lemma 8. *Let δ_1 be the unique real root in $(1, 5/4)$ of the irreducible cubic $\delta^3 + 3\delta^2 - \delta - 4 = 0$. Then the equation $\delta/2 = f_{2a}^{\max}(\delta)$ has three real solutions in $(1, 2)$: $\delta = \delta_1$, $\delta = 4/3$, and an extraneous solution at $\delta = 1$ arising from the rationalisation; the cubic $\delta^3 + 3\delta^2 - \delta - 4$ has its other two real roots in $\mathbb{R}_{<0}$, hence not in $(1, 2)$.*

Proof. Multiplying the equation $\delta/2 = f_{2a}^{\max}(\delta)$ through and substituting $s := \sqrt{\delta}$, $t := \sqrt{\delta - 1}$ collects every non-rational term into a single product $s^j t$; rearranging gives an identity of the form $A(s) = B(s)t$, and squaring (using $t^2 = s^2 - 1$) yields a polynomial identity in s alone. After substituting $s^2 = \delta$, the resulting degree-5 polynomial factorises (over $\mathbb{Q}$) as

$$-\frac{1}{4}(\delta - 1)(3\delta - 4)(\delta^3 + 3\delta^2 - \delta - 4). \quad (12)$$

The linear factors give $\delta = 1$ (extraneous, an artefact of the squaring) and $\delta = 4/3$. The cubic factor has discriminant

$$\text{disc}(\delta^3 + 3\delta^2 - \delta - 4) = 229 > 0,$$

so it has three distinct real roots; numerical evaluation gives $\delta_1 \approx 1.1149075414767557$, $\delta_2 \approx -1.2541016883650524$, and $\delta_3 \approx -2.8608058531117035$. Irreducibility follows from the rational-root test: the only rational candidates $\pm 1, \pm 2, \pm 4$ are checked not to be roots. $\square$

Theorem 9. F attains its global minimum on $(1, \infty)$ at $\delta = \delta_1$, with value

$$F^* := F(\delta_1) = \frac{\delta_1}{2}.$$

Proof. By ??, $F(\delta) = \max(\delta/2, f_{2a}^{\max}(\delta))$. The function $\delta/2$ is strictly increasing on $\delta > 1$. The function $f_{2a}^{\max}(\delta)$ is U-shaped on $\delta > 1$: it tends to 1 as $\delta \rightarrow 1^+$, is strictly decreasing on $(1, \delta_{\text{opt}})$ for some $\delta_{\text{opt}} > 9/8$, and strictly increasing on $(\delta_{\text{opt}}, \infty)$ (this last statement is a one-variable calculus exercise on $f_{2a}^{\max}(\delta)$ but is most economically verified numerically, see ??).

For $\delta \in (1, \delta_1)$, $f_{2a}^{\max}(\delta) > \delta/2$, so $F(\delta) = f_{2a}^{\max}(\delta)$ is decreasing. For $\delta > \delta_1$, $f_{2a}^{\max}(\delta) \leq \delta/2$ until the next intersection point at $\delta = 4/3$ (?); on this whole interval $F(\delta) = \delta/2$ is increasing. Hence the global minimum of F on $(1, \infty)$ is at the first intersection point $\delta = \delta_1$, with $F^* = \delta_1/2$. $\square$

Proof of ??. By ??, $F^* = \delta_1/2 < 9/16$. Substituting $\delta = 2x$ in the cubic of ?? gives

$$(2x)^3 + 3(2x)^2 - (2x) - 4 = 8x^3 + 12x^2 - 2x - 4 = 2(4x^3 + 6x^2 - x - 2), \quad (13)$$

so $F^* = \delta_1/2$ is the unique real root of $4x^3 + 6x^2 - x - 2 = 0$ in $(\frac{1}{2}, \frac{5}{8})$. To put δ_1 in trigonometric form, substitute $\delta = y - 1$ in the cubic of ??; this yields the depressed cubic

$$y^3 - 4y - 1 = 0. \quad (14)$$

With $p = -4$, $q = -1$, the discriminant is $-4p^3 - 27q^2 = 256 - 27 = 229 > 0$, so all three roots are real; Cardano's trigonometric formula gives

$$y_k = \frac{4}{\sqrt{3}} \cos\left(\frac{1}{3} \arccos \frac{3\sqrt{3}}{16} - \frac{2\pi k}{3}\right), \quad k \in \{0, 1, 2\}.$$

The principal value $y_0 = (4/\sqrt{3}) \cos((1/3) \arccos(3\sqrt{3}/16))$ satisfies $y_0 \approx 2.115$, hence $\delta_1 = y_0 - 1 \approx 1.115 \in (1, 5/4)$ as desired, and $F^* = \delta_1/2$ takes the closed form stated in ??.

The remainder of the proof of (??) – specifically Propositions 3.3 and 3.4 of [?] and the supporting Claims 3.5 and 3.6 – is robust to the replacement $\delta = 9/8 \rightsquigarrow \delta_1$, as verified in ??. Hence Lemma 2.3 of [?] holds with c_{FPS} replaced by F^* . $\square$

4 Robustness of [?], Propositions 3.3 and 3.4

The proof of (??) uses, beyond Claim 3.7 itself, the following auxiliary lemmas of [?]:

- Claim 3.5 ([?, p. 8]): upper bound on $|U_w|$ given ℓ_w, ℓ .
- Claim 3.6 ([?, p. 8]): high-probability lower bound on $\delta \cdot k - |U_w|$.
- Proposition 3.3 ([?, p. 8]): $\Delta(H) \leq (F(\delta) + o(1))k$.
- Proposition 3.4 ([?, p. 11]): $\mu(H) = o(k)$.

?? is essentially a restatement of Proposition 3.3 with δ free, so the latter holds at $\delta = \delta_1$ by construction. We check the remaining three.

Claim 3.5. Reads: $|U_w| \leq \ell_w + (d - \ell_w)(k - \ell)/(2k - \ell - 1) + o(k)$. Its proof is a counting argument that uses no numerical specialisation of d/k beyond $d \geq \ell + 1$, which is satisfied for any $\delta \in (1, 5/4)$.

Claim 3.6. States that the random part of U contributes at most a certain expected number of additional U -neighbours to w , with an error $o(k)$. Reading [?, p. 9]: “Set $4d < 5k$.” This is the only place in the proof of Claim 3.6 where $\delta = d/k$ is explicitly constrained, requiring $\delta < 5/4$. Since $\delta_1 \approx 1.115 < 5/4 = 1.25$, Claim 3.6 holds at $\delta = \delta_1$ with the same proof.

Proposition 3.4. The proof ([?, p. 11]) uses $d = \delta k$ in the single quantitative bound

$$\#\{w \in W : w \text{ has } \geq \varphi(k) \text{ neighbours in } U\} \leq \frac{kd}{\varphi(k)} \leq 2k^{1.1},$$

i.e. the assertion $\delta \leq 2$. This is trivially satisfied at $\delta_1 \approx 1.115$. The rest of the Proposition 3.4 proof uses δ only through the cap $|U(d)| \leq k - \varphi(k)$ in [?, Footnote 1, p. 6], which is a structural choice independent of the numerical value of δ .

The aggregate admissible range for δ across all of these inputs is $\delta \in (1, 5/4)$, and $\delta_1 = 1.114907\dots$ lies in the interior of this interval.

5 Numerical verification and reproducibility

We verified the calculations of ?? symbolically using SymPy 1.14. The script reproduces $F(9/8) = 9/16$, $f_{2a}^{\max}(9/8) = 11/20$, $\alpha^*(\eta, 9/8) = (9 - 3/\eta)/8$, $\eta_b(9/8) = 5/7$ exactly, and computes the rationalised polynomial in (??), the discriminant 229 of the cubic, and the roots of the cubic and of $4x^3 + 6x^2 - x - 2$ to 30 significant figures. A second script verifies that the FPS-printed Case 2b objective $(7\eta + 1)/8 - (2 + 1/\eta)/(8 - 7\eta)$ ([?, p. 10]) appears to contain a sign typo: replacing the $+$ by $-$ in the numerator of the second term gives exactly the value $9/16$ at $\eta = 1/2$, while the printed form does not.

Remark 10 (Sign typo in [?, p. 10]). The change $(2 + 1/\eta) \rightsquigarrow (2 - 1/\eta)$ in the second term of the Case 2b objective is consistent with the surrounding algebra ($\alpha = 2 - 1/\eta$, $\delta = 9/8$, hence $\alpha(\delta - 1) = (2 - 1/\eta)/8$) and gives the claimed value $9/16$ at $\eta = 1/2$. The printed sign gives a negative value at $\eta = 1/2$, contradicting [?, Claim 3.7]. The conclusion of Claim 3.7 is unaffected.

6 Downstream and limitations

Remark 11 (Downstream impact on [?] Theorem 1.2). Theorem 1.2 of [?] states that Albertson’s conjecture (??) holds for every k -chromatic graph on at most $1.4(k - 1)$ vertices, with an asymptotic bound of $(1.64 - o(1))k$. The asymptotic coefficient 1.64 is derived from $c_{\text{FPS}} = 9/16$ via the inequality $\chi'(H_i) \leq k - k_i$. Replacing c_{FPS} by F^* (??) replaces 1.64 by a slightly larger value $g(F^*)$, where g is the explicit function of c_{FPS} given in [?, §2]; the magnitude of the change is on the order of 10^{-2} , i.e. the asymptotic Albertson bound moves from $(1.64 - o(1))k$ to roughly $(1.65 - o(1))k$. We do not redo the bookkeeping here. The improvement leaves the finite Cranston residuals $(t, |G|) \in \{(25, 48), (26, 50), (26, 51)\}$ [?] untouched.

Remark 12 (Sharpness of F^*). F^* is sharp *within* the three-case framework of [?, Claim 3.7], the Vizing–Gupta bound (??), and the semi-random construction of [?, §3]. Several routes might yield further improvement:

1. Replacing Vizing–Gupta in (??) by a stronger chromatic-index estimate. For instance, the Goldberg–Seymour theorem ([?] for the proof) bounds $\chi'(H)$ by $\max(\Delta(H), \lceil \Gamma(H) \rceil)$, where $\Gamma(H)$ is the fractional chromatic index; replacing $\Delta + \mu$ by $\max(\Delta, \lceil \Gamma \rceil)$ removes the μ contribution entirely if $\Gamma \leq \Delta$, which is conjectural but plausible for the semi-random multigraphs constructed in [?].

2. Refining the case split in [?, Claim 3.7]. The Case-2b binding inequality reduces to the elementary estimate $|U_w| \leq d/2$ when the high-degree set $U(d)$ is empty; this is tight under uniform random selection and cannot be improved without changing the random selection itself.
3. Changing the semi-random construction in [?, §3] so that the threshold $d = \delta k$ is replaced by a non-constant or adaptive choice.

Each of these is a separate research direction.

Acknowledgements

This note is a small follow-up to the work of Fox, Pach and Suk [?] and would not exist without their construction. Computer-algebra verification was performed in SymPy 1.14.

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